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[MariaDB [projects] > create database Projects;

[MariaDB [projects] > create table Employee(empNo varchar(6), fname
varchar(15), lname varchar(15), address varchar(30), DOB date,sex
varchar(5),
position varchar(15), deptNo varchar(5), primary key(empNo), check (sex
in ("M","F")),
check (position in ("Manager","Team Leader", "Analyst","Software
developer")), foreign key (deptNo) references department(deptNo));

[MariaDB [projects] > create table department(deptNo varchar(5),
deptName varchar(10),mgrEmpNo varchar(5), primary key (deptNo));
alter table department modify column mgrEmpNo varchar(5), add
constraint foreign key(mgrEmpNo) references Employee(empNo);
show columns from department;

[MariaDB [projects] > create table project (projNo varchar(5),projName
varchar(10), deptNo varchar(5), primary key(projNo), foreign key (deptNo)
references department (deptNo));

[MariaDB [projects] > create table worksOn (empNo varchar (5), projNo
varchar(5), dateWorked date, hoursworked int, primary key
(empNo,projNo,dateworked),
foreign key(empNo) references employee (empNo), foreign key (projNo)
references project (projNo), check (hoursWorked between 0 and 40));
show columns from workson;

[MariaDB [projects]> INSERT INTO Department (deptNo, deptName)
VALUES

('D001', 'IT'),
('D002', 'Finance'),
('D003', 'HR'),
('D004', 'Marketing'),
('D005', 'Planning');

[MariaDB [projects] > INSERT INTO Employee (empNo, fName, lName,
address, DOB, sex, position, deptNo) VALUES

('E001', 'Alice', 'Brown', 'Accra', '1985-03-12', 'F', 'Manager', 'D001'),
('E002', 'John', 'Smith', 'Kumasi', '1982-06-20', 'M', 'Manager', 'D002'),
('E003', 'Grace', 'Taylor', 'Cape Coast', '1988-11-05', 'F', 'Manager', 'D003'),
('E004', 'Peter', 'Johnson', 'Tamale', '1984-01-18', 'M', 'Manager', 'D004'),
('E005', 'Linda', 'Wilson', 'Takoradi', '1987-09-09', 'F', 'Manager', 'D005'),
('E006', 'David', 'Owusu', 'Accra', '1992-04-14', 'M', 'Analyst', 'D001'),
('E007', 'Sarah', 'Mensah', 'Kumasi', '1994-07-22', 'F', 'Software Developer',
'D001'),
('E008', 'Michael', 'Asare', 'Sunyani', '1993-08-30', 'M', 'Team Leader',
'D002'),
('E009', 'Ruth', 'Boateng', 'Ho', '1995-12-17', 'F', 'Analyst', 'D003'),
('E010', 'James', 'Adjei', 'Tema', '1991-05-28', 'M', 'Software Developer',
'D005');

```
+-----+-----+-----+-----+-----+-----+-----+
| empNo | fName | lName | address | DOB      | sex | position | deptNo |
+-----+-----+-----+-----+-----+-----+-----+
| E001  | Alice | Brown | Accra   | 1985-03-12 | F   | Manager  | D001   |
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E002 John Smith Kumasi 1982-06-20 M Manager D002
E003 Grace Taylor Cape Coast 1988-11-05 F Manager D003
E004 Peter Johnson Tamale 1984-01-18 M Manager D004
E005 Linda Wilson Takoradi 1987-09-09 F Manager D005
E006 David Owusu Accra 1992-04-14 M Analyst D001
E007 Sarah Mensah Kumasi 1994-07-22 F Software Developer D001
E008 Michael Asare Sunyani 1993-08-30 M Team Leader D002
E009 Ruth Boateng Ho 1995-12-17 F Analyst D003
E010 James Adjei Tema 1991-05-28 M Software Developer D005
+-----+-----+-----+-----+-----+-----+-----+

```

update department set mgrEmpNo ="E001" where deptNo="D001";
update department set mgrEmpNo ="E002" where deptNo="D002";
update department set mgrEmpNo ="E003" where deptNo="D003";
update department set mgrEmpNo ="E004" where deptNo="D004";
update department set mgrEmpNo ="E005" where deptNo="D005";

```

+-----+-----+-----+
deptNo deptName mgrEmpNo
+-----+-----+-----+
D001 IT E001
D002 Finance E002
D003 HR E003
D004 Marketing E004
D005 Planning E005
+-----+-----+-----+

```

[MariaDB [projects] > INSERT INTO Project (projNo, projName, deptNo)
VALUES

```

('P001', 'Student System', 'D001'),
('P002', 'Payroll', 'D002'),
('P003', 'Recruitment', 'D003'),
('P004', 'Website', 'D001'),
('P005', 'Inventory', 'D002'),
('P006', 'Training', 'D003'),
('P007', 'Advertising', 'D004'),
('P008', 'Budget', 'D002'),
('P009', 'Research', 'D005'),
('P010', 'CRM', 'D004');

```
+-----+-----+-----+
| projNo | projName      | deptNo |
+-----+-----+-----+
| P001   | Student System | D001   |
| P002   | Payroll        | D002   |
| P003   | Recruitment     | D003   |
| P004   | Website         | D001   |
| P005   | Inventory       | D002   |
| P006   | Training        | D003   |
| P007   | Advertising     | D004   |
| P008   | Budget          | D002   |
```

P009	Research	D005	
P010	CRM	D004	
+-----+-----+-----+			

[MariaDB [projects] > INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES

('E001', 'P001', '2026-06-01', 8),
 ('E006', 'P001', '2026-06-01', 7),
 ('E007', 'P004', '2026-06-01', 6),
 ('E002', 'P002', '2026-06-02', 8),
 ('E008', 'P005', '2026-06-02', 5),
 ('E003', 'P003', '2026-06-03', 7),
 ('E009', 'P006', '2026-06-03', 8),
 ('E004', 'P007', '2026-06-04', 6),
 ('E005', 'P009', '2026-06-05', 8),
 ('E010', 'P009', '2026-06-05', 7);

+-----+-----+-----+-----+

empNo	projNo	dateWorked	hoursworked	
-------	--------	------------	-------------	--

+-----+-----+-----+-----+

E001	P001	2026-06-01	8	
E002	P002	2026-06-02	8	
E003	P003	2026-06-03	7	
E004	P007	2026-06-04	6	

E005 P009 2026-06-05 8
E006 P001 2026-06-01 7
E007 P004 2026-06-01 6
E008 P005 2026-06-02 5
E009 P006 2026-06-03 8
E010 P009 2026-06-05 7

+-----+-----+-----+-----+

```
[MariaDB [projects] > create view Female_Projects as select
projNo,projName fFROM Project P JOIN Department D ON P.deptNo =
D.deptNo JOIN Employee E ON D.mgrEmpNo = E.empNo WHERE E.sex =
'F' AND E.position = 'Manager' ORDER BY P.projNo;
```

```
[MariaDB [projects] > create view Employee_Details as select
employee.empNo, fname,lname,projName, hoursworked from employee
join worksON on employee.empNo=workson.empNo join project on
workson.projNo=project.projNo;
```

```
[MariaDB [projects] > CREATE VIEW EmpProject(empNo, projNo,
totalHours) AS SELECT employee.empNo, project.projNo,
SUM(hoursWorked) from Employee, Project, WorksOn where
employee.empNo = workson.empNo and project.projNo = workson.projNo
group by workson.empNo, workson.projNo;
```

```
[MariaDB [projects] > Select * from emproject;
```

+-----+-----+-----+

empNo projNo totalHours

+-----+-----+-----+

E001 P001 8

E002 P002 8

E003 P003	7
E004 P007	6
E005 P009	8
E006 P001	7
E007 P004	6
E008 P005	5
E009 P006	8
E010 P009	7
+-----+-----+-----+	

[MariaDB [projects] > select * projNo form empProject where projNo="P001";

+-----+
projno
+-----+
P001
P001
+-----+

[MariaDB [projects] > Select Count (projNo) form empproject where empNo = "E001";

+-----+
count(projno)
+-----+
1
+-----+

[MariaDB [projects] > select empNo, totalHours from empProject group by empNo;

```
+-----+-----+
| EMPNO | totalhours |
+-----+-----+
| E001  |      8 |
| E002  |      8 |
| E003  |      7 |
| E004  |      6 |
| E005  |      8 |
| E006  |      7 |
| E007  |      6 |
| E008  |      5 |
| E009  |      8 |
| E010  |      7 |
+-----+-----+
```

- Insert: If a new part has partCost > 1000 then the partNo should be added to the materialized view. Otherwise we don't edit the view.
 - Update: If a part's cost changes from 1000 or below to above 1000, add it to the view. If its cost changes from above 1000 to 1000 or below, remove it from the view. If the part remains on the same then no change is needed.
 - Delete: If an expensive part is deleted, it should be removed from the materialized view. If a non-expensive part is deleted nothing should be done.
-
- The database does not need to check the Part table if one row is enough to decide whether to add or remove a part from the view.
 - If the same partNo appears more than once, it must check the Part table to make sure there are no other expensive records for that part

